```python
# Pick the dad haplotype that was transmitted to the kid.
df_meth = df_meth.with_columns(
    dad_transmitted = pl.when(pl.col("pat_hap") == "A")
        .then(pl.col("dad_A"))
        .otherwise(pl.col("dad_B"))
)

# Find runs of consecutive CpGs where the kid_pat is methylated
# but the dad-transmitted haplotype is not - i.e., a de novo
# *gain* of methylation on the kid paternal homolog.
WINDOW = 2
df_denovo_gain_pat = (
    df_meth.with_columns(
        delta = pl.col("kid_pat") - pl.col("dad_transmitted"),
    )
    .with_columns(
        mean_delta = pl.col("delta").rolling_mean(WINDOW),
    )
    .filter(pl.col("mean_delta") > 0.5)
)
```